LESSON 4.10
APPLYING MULTIPLE LAWS OF EXPONENTS – PART 1

LESSON INTRODUCTION

MA.8.AR.1.1 Apply the laws of exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

LEARNING TARGETS

- I can write an equivalent algebraic expression by applying multiple laws of exponents.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational bases.

WORKING TOWARD

MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.

ESSENTIAL QUESTIONS

- Does order matter when applying the laws of exponents to generate an equivalent algebraic expression?
- How did you determine which order to apply the laws of exponents when generating an equivalent algebraic expression?

LESSON NARRATIVE

This lesson is a culmination of the previous lessons in the unit. Students apply multiple exponent laws to generate equivalent algebraic expressions. Students analyze multiple student work samples to recognize that sometimes the same laws are applied just in a different order but still yield the same equivalent expression. Students will also see that sometimes the order in which laws are applied eliminates the need to apply other laws.

COMPONENT SUMMARY

4.10.1 WARM-UP *(~5 MIN.)*

Complete a number puzzle

4.10.2 GUIDED PRACTICE *(15 MIN.)*

Analyze student work where multiple exponent laws are used

4.10.3 TRY IT *(5-10 MIN.)*

Apply multiple laws of exponents to generate equivalent algebraic expressions

4.10.4 ACTIVITY *(15 MIN.)*

Generate equivalent algebraic expressions in a relay activity

4.10.5 YOUR TURN! *(5-10 MIN.)*

Apply multiple laws of exponents to generate equivalent algebraic expressions

4.10.6 WRAP-UP *(~5 MIN.)*

Demonstrate mastery of learning target by writing an equivalent expression

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse the operations applied to the exponents and coefficients. For example, when applying the power of products, they may add coefficients or multiply exponents. When applying the quotient of powers law, they may subtract the coefficients or divide the exponents.

THINGS TO CONSIDER

- Incorrectly answering any step in the 4.10.4 Activity will result in an incorrect final answer. If each partner does not recognize the mistake(s), prompt them to check their work with another pair of students.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

4.10.2 Guided Practice prompts students to share what they notice and wonder about each student's work. Ask them to discuss the exponent laws that were applied at each step and why they did or did not result in equivalent expressions during each step.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Students demonstrate fluency and/or build fluency (for those who are still not fluent) throughout the entire lesson. Students have opportunities to be flexible in their choice of which exponent law they apply as well as revise their thinking during the 4.10.4 Activity. If students do not have the result of 1 at the end of the activity, they must determine where error(s) were made in their work and revise the steps they took to write equivalent expressions.

DIFFERENTIATE INSTRUCTION

For students who are struggling with mathematical fluency, consider having them write expressions in expanded form to generate an equivalent expression. Have students do this until they feel comfortable not using expanded form. This will give students the choice and flexibility to use the additional support (additional step) until they feel confident enough to move past it.

Allow students to refer to prior lessons, their reference sheet, or their anchor chart for examples of applying the different exponent laws.

Additional differentiation opportunities are denoted throughout the lesson guide.

4.10.1 WARM-UP: CRACK THE SAFE

DIFFERENTIATE INSTRUCTION

If students are lost, allow them to work in pairs to complete the 4.10.1 Warm-Up. Productive struggle is good for students, but if students are lost to where they are shutting down at the start of the lesson, allow them the opportunity to work in pairs.

4.10.1 Warm-Up: Crack the Safe

1. To crack the safe, you must fill in the middle columns with numbers from the bank. Each number can only be used once. Each row must add to equal 18. You have 5 minutes before it is locked forever.

Number Bank									
8	4	5	2	9	6	7	13	1	3

Code			Row Total
3	9	6	18
6	7	5	18
2	13	3	18
9	1	8	18
12	2	4	18

Answers may vary

4.10.2 Guided Practice: Applying Multiple Laws of Exponents to Algebraic Expressions

The laws of exponents are included in the bank below for you to reference as you complete the following problems.

Laws of Exponents

Product of Powers	Quotient of Powers	Power of a Power
Power of a Product	Power of a Quotient	Negative Exponent
Identity Exponent	Zero Exponent	

1. Miguel and Gia generated the same expression equivalent to $(r^3 \cdot r^2)^2$. They both applied laws of exponents, but in different ways. Their work is shown.

Miguel	Gia
$(r^3 \cdot r^2)^2$	$(r^3 \cdot r^2)^2$
$(r^5)^2$	$(r^6 \cdot r^4)$
r^{10}	r^{10}

   a. For each student, identify which laws of exponents they applied and in what order.

Order of Laws Applied	Miguel	Gia
First	*Product of Powers*	*Power of a Product*
Second	*Power of a Power*	*Product of Powers*

   b. Discuss with your partner which strategy you prefer. Summarize your discussion.
   Answers will vary. Encourage students to explain why they prefer a particular strategy.

2. Jack and Angel rewrote the expression $\left(\frac{x^3y^{-4}}{xy^2}\right)^2$ in different ways and arrived at different expressions. Their work is shown.

4.10.2 GUIDED PRACTICE: APPLYING MULTIPLE LAWS OF EXPONENTS TO ALGEBRAIC EXPRESSIONS

TEACHER MOVES

Notice and Wonder

Prompt students to share what they notice and wonder about the samples of student work. Ask them to discuss the laws of exponents that were applied at each step and why they did or did not result in equivalent expressions.

Jack	Angel
$\left(\frac{x^3y^{-4}}{xy^2}\right)^2$	$\left(\frac{x^3y^{-4}}{xy^2}\right)^2$
$(x^3y^{-2})^2$	$\frac{x^6y^{-8}}{x^2y^4}$
x^6y^{-4}	x^4y^{-12}
$\frac{x^6}{y^4}$	$\frac{x^4}{y^{12}}$

   a. Even though one of the student's resulting expressions is not equivalent to $\left(\frac{x^3y^{-4}}{xy^2}\right)^2$, identify which laws of exponents each student applied and in what order.

Order of Laws Applied	Jack	Angel
First	*Quotient of Powers*	*Power of a Quotient*
Second	*Power of a Product*	*Quotient of Powers*
Third	*Negative Exponent*	*Negative Exponent*

   b. Which student generated an expression that is not equivalent to $\left(\frac{x^3y^{-4}}{xy^2}\right)^2$?
   Jack

   c. Which statement best describes the incorrect student's strategy?
   - The incorrect student used a correct strategy, but applied one of the laws incorrectly.
   - The incorrect student tried to use an exponent law that didn't apply to this expression.

   d. Based on your response to Part C, show how to correct the student's error.

$$\left(\frac{x^3y^{-4}}{xy^2}\right)^2$$
$$(x^2y^{-6})^2$$
$$x^4y^{-12}$$
$$\frac{x^4}{y^{12}}$$

TEACHER MOVES

Angel applied two laws in one step: power of a product and power of a quotient. Most students will likely see power of a quotient, and that is okay. While students are asked to name the laws, the purpose of that is really to see patterns and relationships, not to overemphasize memorizing the names of laws. The names should be given, and ideally there is an anchor chart somewhere in the room to reference.

The goal is for students to notice that sometimes the same laws can be applied in different orders to arrive at the same equivalent expression. Also, students may notice that sometimes the order in which the laws are applied can eliminate the need to apply the other laws

ENRICHMENT

- Which law did Jack apply incorrectly?
- How did you determine the mistake?

4.10.3 TRY IT: APPLYING MULTIPLE LAWS OF EXPONENTS TO ALGEBRAIC EXPRESSIONS

DIFFERENTIATE INSTRUCTION

For students who need additional scaffolding, consider having them write expressions in expanded form as the first step to generating an equivalent expression.

ENRICHMENT

Could you have taken a different order of steps in applying the exponent laws?

4.10.3 Try It: Applying Multiple Laws of Exponents to Algebraic Expressions

1. For each expression, write an equivalent expression using positive exponents and the fewest factors possible. Show your work in the space provided. Then, for each expression, check off the exponent laws applied from the list provided.

a. $\left(\frac{9c^4}{c^{-4}}\right)^2$ $81c^{16}$

Exponent Laws Applied
☐ identity exponent
☐ negative exponent
☐ power of a power
☐ power of a product
☐ power of a quotient
☐ product of powers
☐ quotient of powers
☐ zero exponent

b. $(d^0 \cdot d^5)^{-2}$ $d^{-10} = \frac{1}{d^{10}}$

Exponent Laws Applied
☐ identity exponent
☐ negative exponent
☐ power of a power
☐ power of a product
☐ power of a quotient
☐ product of powers
☐ quotient of powers
☐ zero exponent

Answers could vary

4.10.4 Activity: Exponent Laws Relay

Work with your partner to complete the relay.

- Start by generating an equivalent expression for question 1 using the fewest factors possible.
- Insert the answer from question 1 into the box in question 2. Repeat until you have completed all of the questions in this manner.

1. $\frac{5x^{-3} \cdot x^{-2}}{25x^{-4}} = \frac{1}{5x}$
2. $10x^3 \cdot \boxed{\frac{1}{5x}} = 2x^2$
3. $\frac{(2x^{-4} \cdot x^5)^3}{\boxed{2x^2}} = 4x$
4. $\left(\frac{x}{2x^3}\right)^2 \cdot \boxed{4x} = \frac{1}{x^3}$
5. $x^3 \cdot \boxed{\frac{1}{x^3}} = 1$

To successfully complete the relay, your final answer should be 1. Go back to check your answers if needed.

4.10.4 ACTIVITY: EXPONENT LAWS RELAY

ENRICHMENT

What law of exponents did you apply at each step? Do you agree with your partner's work? Why or why not?

TEACHER MOVES

Take Turns

Prompt students to take turns answering each step of the relay. Have students justify their work by sharing their reasonings and the law of exponents that they used.

4.10.5 YOUR TURN!

QUESTIONING

If students finish early, ask them if they can generate the equivalent expression by applying the laws of exponents in a different order.

4.10.5 Your Turn!

1. For each expression, write an equivalent expression using positive exponents and the fewest factors possible. Show your work in the space provided. Then, for each expression check off the exponent laws applied from the list provided to the right of each expression.

a. $\frac{8x^3}{2x^9} \cdot x^4$ $\frac{4}{x^2}$

Exponent laws applied may vary.

Exponent Laws Applied
☐ identity exponent
☐ negative exponent
☐ power of a power
☐ power of a product
☐ power of a quotient
☐ product of powers
☐ quotient of powers
☐ zero exponent

b. $\left(\frac{p^0 \cdot q^{-2}}{(p^{-3} \cdot q)^2}\right)$ $\frac{p^6}{q^4}$

Exponent laws applied may vary.

Exponent Laws Applied
☐ identity exponent
☐ negative exponent
☐ power of a power
☐ power of a product
☐ power of a quotient
☐ product of powers
☐ quotient of powers
☐ zero exponent

4.10.6 Wrap-Up: Ticket Out the Door

1. Write an equivalent expression using positive exponents and the fewest factors possible.

$$\left(\frac{3m^3n^2}{6mn^4}\right)^2$$

$$\frac{m^4}{4n^4}$$

4.10.6 WRAP-UP: TICKET OUT THE DOOR

QUESTIONING

If students finish early, challenge them to write an expression that requires the application of as many laws of exponents as possible to generate an equivalent expression using positive exponents and the fewest factors possible.

DIFFERENTIATE INSTRUCTION

Consider using student responses from prior self-reflections, along with their process and answer to the 4.10.6 Wrap-Up, to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.